

ESP32-S3 Plant Monitor — Bring-Up Checklist

Rev A · first power-on, in order · tick each box at the bench

Board ID / serial: _____

Date: _____

Tester: _____

⚠ **Before you start:** use a current-limited USB source if you have one, and **meter the battery plug polarity** before it ever touches J3. Q1 makes a reversed battery survivable — the meter makes it a non-event.

A · Before power (nothing plugged in)

☐	1. Visual inspection against the schematic: orientation of U1-U6, D1-D4, Q1-Q3; polarity marks; no solder bridges.	
☐	2. Meter, no power: confirm NO short between GND and each of +5V_PROT, VSYS, +3V3, VBAT.	All open / high resistance

B · USB power (current-limited if possible)

☐	3. Apply USB-C. Check TP1 (+5V_PROT).	≈ 5 V
☐	4. Check TP14 (VSYS).	≈ 4.6 V
☐	5. Check TP2 (+3V3). Power LED D1 should be on.	3.3 V
☐	6. Confirm the board enumerates as a USB serial device; flash a blink / hello firmware.	Appears + flashes
☐	7. Run an I ² C scan.	Devices at 0x76 and 0x10

C · Battery & automatic power path

☐	8. Meter battery polarity FIRST, then plug battery into J3 (USB still in). Charge LED D2 on; TP11 climbs; charge current.	TP11 → 4.2 V, ≈ 100 mA
☐	9. Hand-over test: board running, meter/scope on TP2 — unplug USB. Board must keep running.	TP14 → battery V; TP2 stays 3.3 V
☐	10. Battery-only boot: power up from the battery alone (no USB).	Boots normally

D · Sensors, calibration & soak

☐	11. Buzz out the soil-probe cable with a meter, then plug into J2. Record dry and in-water readings; compare to breadboard constants.	Dry / wet logged
☐	12. Sleep-current measurement in series with the battery (unsolder D1 for a clean figure).	≈ 100 μA floor
☐	13. Wi-Fi range sanity check, then a 24-hour soak run.	Stable over 24 h

Success criteria (from the design doc): powers up on USB and accepts firmware · all sensors read, soil calibrated · unplugging USB does not reset the board · charges at ~100 mA · deep-sleep current low enough for multi-week battery life.

Source: ESP32-S3 Plant Monitor Final Design Document, Rev A — Sections 11 & 13.